$\beta_0 = 0.01635$
$m_0 = 3.7426$
$b_0 = 54.8732$

FA23-BCE-113
Muhammad Ahmad

$\hat{y} = m_0\hat{x} + \hat{b}$

T+1 Date:

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$X[Y-\hat{Y}]$
2	69	62.3584	6.6416	13.2832
9	98	88.5566	9.4434	84.9906
5	82	73.586	8.714	43.57
5	77	73.586	+3.414	+17.07
3	71	66.101	4.899	14.697
7	84	81.0714	2.926	20.482
1	55	58.6155	-3.6155	-3.6155
8	94	84.814	9.186	73.488
6	84	77.3288	6.6712	40.0272
2	64	62.3584	1.6416	3.2832
		Σ	49.6232	305.7916

① grad $m = \frac{-2}{n} \times \sum_{i=1} \hat{x}(y-\hat{y})$

$= \frac{-2}{10} \times 305.7916$

grad $m = 61.15832$

② grad $b = \frac{-2}{n} \sum_{i=1} (y-\hat{y})$

$= \frac{-2}{n} \times 49.6232$

grad $b = -9.9246$

Date:

M T W T F S

$\Rightarrow m_1 = m_0 - \lambda \text{ grad } m$

$= 3.7426 - (0.01635)(-61.1583)$

$m_1 = 4.742539$

$\Rightarrow b_1 = b_0 - \lambda \text{ grad } b$

$= 54.8732 - (0.01635)(-9.92464)$

$b_1 = 55.03548$

now

$$\boxed{m_1 = 4.742539, \quad b_1 = 55.03548}$$

IT 2

$\hat{y}$	$y-\hat{y}$	$x[y-\hat{y}]$
64.5205	4.4795	8.9589
97.7183	0.02817	2.5352
78.7482	3.2518	16.2592
78.7482	-1.7482	-8.7408
59.2631	1.7369	5.2107
88.2332	-4.2332	-29.627
59.7780	-4.7780	-4.7780
92.9758	1.0242	8.19.39
83.4907	0.5093	3.0558
64.5205	-0.5205	-1.0411
Σ	0.0035	0.0211

Date:

(M)(T)(W)(T)(F)(S)

① grad m:

$$= \frac{-2}{10} \times 0.0211$$

$$= -0.00422$$

② grad b

$$= \frac{-2}{10} \times 0.0035$$

$$= -0.0007$$

$\Rightarrow m_2 = 4.74239 - 0.01635(-0.00422)$

$m_2 = 4.7426$

$\Rightarrow b_2 = 55.035468 - 0.1635(-0.0007)$

$b_2 = 55.0354$

	m	b
0	3.74	54.8732
1	4.742539	55.03548
2	4.7426	55.0354